SME Technology Directory

Version 2023-12-12

		Improvement		Reduction					Improvement			Financial benefits	
#	Module	Code	Description	Energy	Water	Waste	Raw material	Carbon footprint	TRL	Prod. efficiency	Methods	Cost red.	Income incr.
1	BM1	BM1-01	Renewable energy (Wind energy, Solar energy, ...)	X				X				X	X
2	BM1	BM1-02	Energy-efficient technologies (LED lighting with sensors, Efficient consumer electronics, ...)	X	X	X		X				X	X
3	BM1	BM1-03	Sustainable supply chains	X			X	X		X		X	X
4	BM1	BM1-04	Social impact initiatives					X					X
5	BM2	BM2-01	Sustainability transition strategic planning					X		X		X	X
6	BM2	BM2-02	Integration of sustainable strategic planning	X	X	X	X	X		X		X	X
7	BM3	BM3-01	Competitive advantage (technology and organization)	X		X		X		X	X	X	X
8	BM3	BM3-02	Cost saving (more effective technologies)	X						X	X	X	X
9	BM3	BM3-03	Regulatory compliance	N/A	N/A	N/A	N/A	N/A			X	X	X
10	BM3	BM3-04	Market access and reputation										X
11	BM3	BM3-05	Long term viability									X	X
12	BM4	BM4-01	Financial planning									X	X
13	BM4	BM4-02	Capital budgeting									X	X
14	BM4	BM4-03	Investment decisions									X	X
15	BM7	BM7-01	Change leadership									X	X
16	BM7	BM7-02	Change Readiness Assessment									X	X
17	BM7	BM7-03	Change Communication									X	X
18	BM7	BM7-04	Change Planning and Implementation	X	X	X	X	X				X	X
19	BM7	BM7-05	Change Engagement and Support									X	X
20	BM7	BM7-06	Change Evaluation and Continuous Improvement									X	X
21	PA2	PA2-01	Advanced Robotics	X					X	X	X	X	X
22	PA2	PA2-02	Advanced Cobotics	X					X	X	X	X	X
23	PA5	PA5-01	Simulation, explanation of Smart Logistics Technologies and software like AGV, AMR, ACR, Smart Warehouse systems, etc.	X		X	X	X		X	Simulation	X	X
24	PA6	PA6-01	VR environment - not real production	X		X	X	X		X	VR	X	X
25	PA6	PA6-02	AR environment - not real production	X		X	X	X		X	AR	X	X
26	PA7	PA7-01	Implementation of edge computing technology							X	X	X	
27	PA7	PA7-02	Synergy with 5g networks								X		
28	PA7	PA7-03	Synergy with cloud(s)	X						X	X	X	
29	PA7	PA7-04	Application of artificial intelligence							X	X	X	X
30	PE2	PE2-01	Material Extrusion	X	X	X	X	X	6	X	FDM,FFF,CFF	X	X
31	PE2	PE2-02	Vat Photopolymerization	X	X	X	X	X	7	X	SLA,DLP	X	X
32	PE2	PE2-03	Powder Bed Fusion	X	X	X	X	X	8	X	SLS,SLM, EBM	X	X
33	PE2	PE2-04	Material Jetting	X	X	X	X	X	6	X	PJM, DOD	X	X
34	PE2	PE2-05	Binder Jetting	X	X	X	X	X	6	X	BJ, MBJ	X	X
35	PE2	PE2-06	Directed Energy Deposition	X	X	X	X	X	6	X	LBAM,WAAM	X	X
36	PE2	PE2-07	Sheet Lamination	X	X	X	X	X	6	X	LOM	X	X
37	PE3	PE3-01	Circular design	X		X		X		X		X	X

38	PE3	PE3-02	Sustainable materials (natural materials, renewable materials, non-toxic and harmless material)	X	X	X	X	X					
39	PE3	PE3-03	Sustainable manufacturing (avoiding causing harm to the local environment)	X	X	X	X	X		X		X	X
40	PE3	PE3-04	Sustainable packaging and transportation (Biodegradable Packaging, Recyclable Packaging, Reusable Packaging, Plant-Based Packaging, ...)	X	X	X	X	X		X		X	X
41	PE3	PE3-05	Next generation materials (higher-performance materials)	X		X	X	X		X			X
42	PS4	PS4-01	Circulate product and materials (repair, reuse, refurbish, recycle)	x	x	x	x	x			x	x	x
43	PS4	PS4-02	Renewable energy	x				x			x		
44	PS4	PS4-03	Reduce waste and pollution			x					x	x	x
45	PS4	PS4-04	Sharing and exchange economy (change of customer behaviour)	x	x	x	x	x				x	x
46	PS5	PS5-01	Redesign the packaging strategy (size, volume, new materials, reduce material usage)	X		X	X	X			X	X	X
47	PS5	PS5-02	Improve the product and package logistic	X				X			X		X
48	PS5	PS5-03	Reuse and recycle the package materials			X	X	X					
49	BM6	BM6-01	Digital technologies - semiconductor components and systems	X					X				X
50	BM6	BM6-02	Digital technologies - photonics	X					X			X	
51	BM6	BM6-03	Digital technologies - cyber-physical systems						X				X
52	BM6	BM6-04	Advanced digital tools for remote team work & management	X							X		X
53	BM8	BM8-01	CSRD								x		
54	BM8	BM8-02	Sustainable reporting to attract sustainable finance								x		x
55	PS2	PS2-01	Critical raw materials				x	x		x	x	x	
56	PS2	PS2-02	Remanufacturing	x		x	x	x		x	x		
57	PS6	PS6-01	Photovoltaics	X				X				X	
58	PS6	PS6-02	Heat pumps	X				X				X	
59	PS6	PS6-03	Water recovery system	X	X			X				X	
60	PS6	PS6-04	Local energy storage	X				X				X	
61	PS9	PS9-01	Sustainability and strategic planning					X			X		
62	PS9	PS9-02	Risk management					X			X		
63	PS9	PS9-03	Carbon footprint calculation					X			X		
64	PS1	PS1-01	Renewable Energy: Implementing solar, wind, or hydropower sources	X				X		X		X	X
65	PS1	PS1-02	Energy-efficient Technologies (Smart sensors, ...)	X	X			X		X		X	X
66	PS1	PS1-03	3D printing			X	X			X			X
67	PE6	PE6-01	Virtual reality technology			X	X	X		X		X	X
68	PE6	PE6-02	Motion Tracking Sensors	X				X		X			
69	PE6	PE6-03	Wireless Connectivity	X						X		X	X
70	PE6	PE6-04	Head-Mounted Displays (HMDs)					X		X			
71	PA3	PA3-01	AI-power predictive maintenance and manufacturing	X	X	X	X	X		X	X	X	X
72	PA3	PA3-02	5G Networks	X				X	X	X	X	X	X
73	PA3	PA3-03	Edge Computing	X				X		X		X	X
74	PE7	PE7-01	Edge Computing	X						X	X	X	X
75	PE7	PE7-02	RFID technology	X	X	X	X	X		X		X	X
76	PE7	PE7-03	Data management and analytics technologies	X			X	X		X	X	X	X

77	PS8	PS8-01	Digital twin	X	X	X	X	X		X		X	X
78	PS8	PS8-02	Simulation	X			X	X		X	X	X	X
79	PS8	PS8-03	Communication protocols	X						X			
80	PS8	PS8-04	Artificial technologies and algorithms	X	X	X	X	X		X	X	X	X
81	PS7	PS7-01	Digital twin	X	X	X	X	X		X		X	X
82	PS7	PS7-02	Internet of Things technologies	X			X	X		X	X	X	X
83	PS7	PS7-03	Edge Computing	X				X		X	X	X	X
84	PA4	PA4-01	Autonomous robots	X						X		X	X
85	PA4	PA4-02	Simulation	X	X	X	X	X		X	X	X	X
86	PA4	PA4-03	Horizontal and vertical system integration	X						X		X	X
87	PA4	PA4-04	The industrial Internet of Things					X		X		X	X
88	PA4	PA4-05	Cybersecurity							X			
89	PA4	PA4-06	Clou computing	X		X	X	X		X		X	X
90	PA4	PA4-07	Additive manufacturing	X	X	X	X	X		X		X	X
91	PA4	PA4-08	Augmented reality	X	X	X	X	X		X	X	X	X
92	PA4	PA4-09	Big data and analytics					X		X	X	X	X
93	PA8	PA8-01	Human simulation				X			X		X	X
94	PA8	PA8-02	Augmented and virtual reality technologies	X	X	X	X	X		X		X	X
95	PA8	PA8-03	Simulation and digital twins	X	X	X	X	X		X		X	X
96	PE5	PE5-01	Bateries	X				X	X	X	X	X	X
97	PE5	PE5-02	Electric vehicle	X				X	X	X	X	X	X
991		OT-00	Other technology										
992		OF-00	Other financial method										
993		OM-00	Other management method										